

Jean-Charles Quillet

SENIOR SOFTWARE ENGINEER · FUNCTIONAL PROGRAMMING

13, rue du clos romain, 35120 Dol-de-Bretagne, FRANCE

☎ (+33) 6-65-13-50-75 | ✉ jeancharles.quillet@gmail.com | 🏠 jeancharles.quillet.org | 📺 jecaro | 🌐 jeancharlesquillet

Summary

Senior software engineer with twenty years of experience across backend systems, network servers, and simulation software. Strong expertise in C++ (10+ years) and functional programming with Haskell (5 years). Proficient in systems design, performance optimization, and developing correct, safe, and robust software. Actively involved in open-source software, both as a contributor and as a maintainer.

Skills

Systems Distributed systems, backend services, simulation software

Methods Test-driven development, code review, CI/CD pipelines

Domains Networking, Computer Graphics, Geographic Information Systems

Programming languages Haskell, Rust, C/C++, Zig, Python, Elm, TypeScript, Bash, Lua, Matlab

Tools Git, nix, PostgreSQL, MySQL, Redis, Kafka, Docker, Kubernetes, OpenTelemetry

Languages French: mother tongue, English: fluent, Spanish: beginner

Experience

Tiko

Zürich, Switzerland / Remote

BACKEND DEVELOPER

Since 01/2024

- Worked on the interface between IoT devices (over 300k deployed) and backend services
- Implemented observability features for an in-house database, improving monitoring, logging, and performance insights
- Participated in on-call rotation for production monitoring and incident handling
- Technical environment: Linux, Haskell, C++, Java, nix, Kubernetes, PostgreSQL, MySQL, Redis, Kafka, MQTT, OpenTelemetry

Secucloud

Hamburg, Germany / Remote

CLOUD SOFTWARE DEVELOPER

11/2020 - 11/2023

- Developed cloud-based network security software as part of a SASE (Secure Access Service Edge) system
- Implemented an Active Directory connector for seamless synchronization of users and groups
- Optimized network proxy performance, increasing throughput by 50%
- Technical environment: Linux, Haskell, TypeScript, nix, MySQL, Ansible, Redis, Clickhouse

Epitech

Rennes, France

FUNCTIONAL PROGRAMMING TEACHER

09/2021 - 06/2023 (part-time)

- Supported students with their Haskell projects
- Technical environment: Linux, Haskell, nix

Scalian

Rennes, France

SOFTWARE DEVELOPER FOR AN IT SERVICES COMPANY

04/2007 - 11/2020

- Developed simulation software within the Simulation, HPC and Virtual Reality department
- Delivered multiple fixed-price projects on time and within budget for clients including the French MoD, ONERA, CEA, CNES, IRSN
- Architecture and technical lead
- Collaborated with researchers and physicists to gather requirements and deliver tailored software solutions
- Technical environment: Windows/Linux, C++, Python, Matlab, Qt, openscenegraph, gdal

LaBRI Analysis and indexing video team (University lab)

Bordeaux, France

RESEARCH ENGINEER

07/2006 - 08/2006

- Prepared the video indexing contests TRECVID and ARGOS
- System administration
- Technical environment: Linux, C/C++, XML, Perl, Bash

INRIA IPARLA team (French IT lab)

Bordeaux, France

RESEARCH ENGINEER

09/2004 - 04/2006

- Visualization of city models on mobile devices through non-photorealistic rendering
- Wrote and presented a scientific paper to the Web3D 2006 conference
- Technical environment: Linux, Windows, Windows Mobile, C/C++, Bash, Perl, VRML97, X3D

Keyghost

SOFTWARE DEVELOPER

- Developed a desktop application to interact with a hardware keylogger
- Technical environment: Windows, Delphi, C/C++, Octave

Christchurch, New-Zealand

04/2003 - 12/2003

Visualpix

INTERNSHIP FOLLOWED BY A SOFTWARE ENGINEER CONTRACT

- Improved a video compression codec by implementing a spatial index
- Technical environment: Linux, C

Bordeaux, France

04/2002 - 12/2002

Glisshouse

BACKEND DEVELOPER, SYSTEM ADMINISTRATOR

- Web: Maintained an online shop, developed a CMS
- System administration: Installed, setup and maintained the LAN
- Technical environment: Linux, Php, MySQL, Apache, Samba, cron

Marseille, France

07/2000 - 03/2001

Education

- 2001 - 2002 **DESS Multimedia and Information Technology**,
image synthesis and analysis, 3D geometry and rendering, sound, video
- 1999 - 2000 **Master in computer science**, image synthesis and analysis
- 1998 - 1999 **B.S. in computer science**
- 1996 - 1997 **DEUG computer science and mathematics**
- 1995 - 1996 **Baccalauréat Science**

Bordeaux university

Poitiers university

Limoges university

Limoges university

Lycée Jean Giraudoux, Châteauroux

Extracurricular Activity

Sports Rock climbing, bike commuting, surf, trekking

Culture Literature (classical, travel, adventure), cinema, attending rock concerts, guitar playing

Community commitments Member of the direction comity of the rock climbing club of Rennes (2012 to 2016)
Volunteered at concert venues of Rennes: Antipode (2011 to 2015), le Jardin Moderne (2009 to 2010)
Volunteered to the music festivals Epipapu (2009, 2010) and la Route du Rock (2008 - 2018)

Travel New-Zealand, Australia, USA, Canada, Malaysia, Bali, Bolivia, Iceland, Norway, Scotland, Faroe Islands